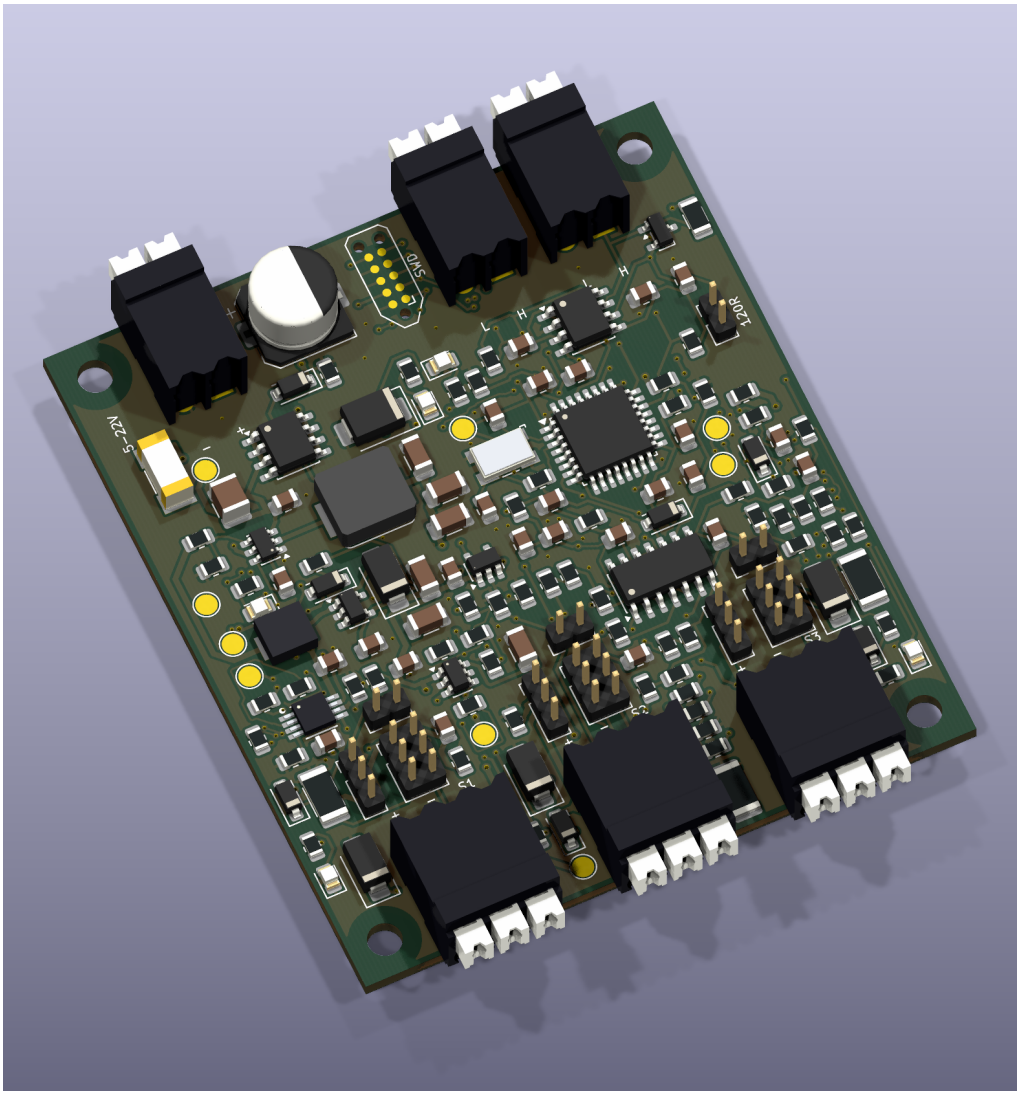
FRC
Sensor CAN

Revision B · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Protected battery and regulated 5 V sensor power



2026-09-08

Editable KiCad files, DigiKey BOM, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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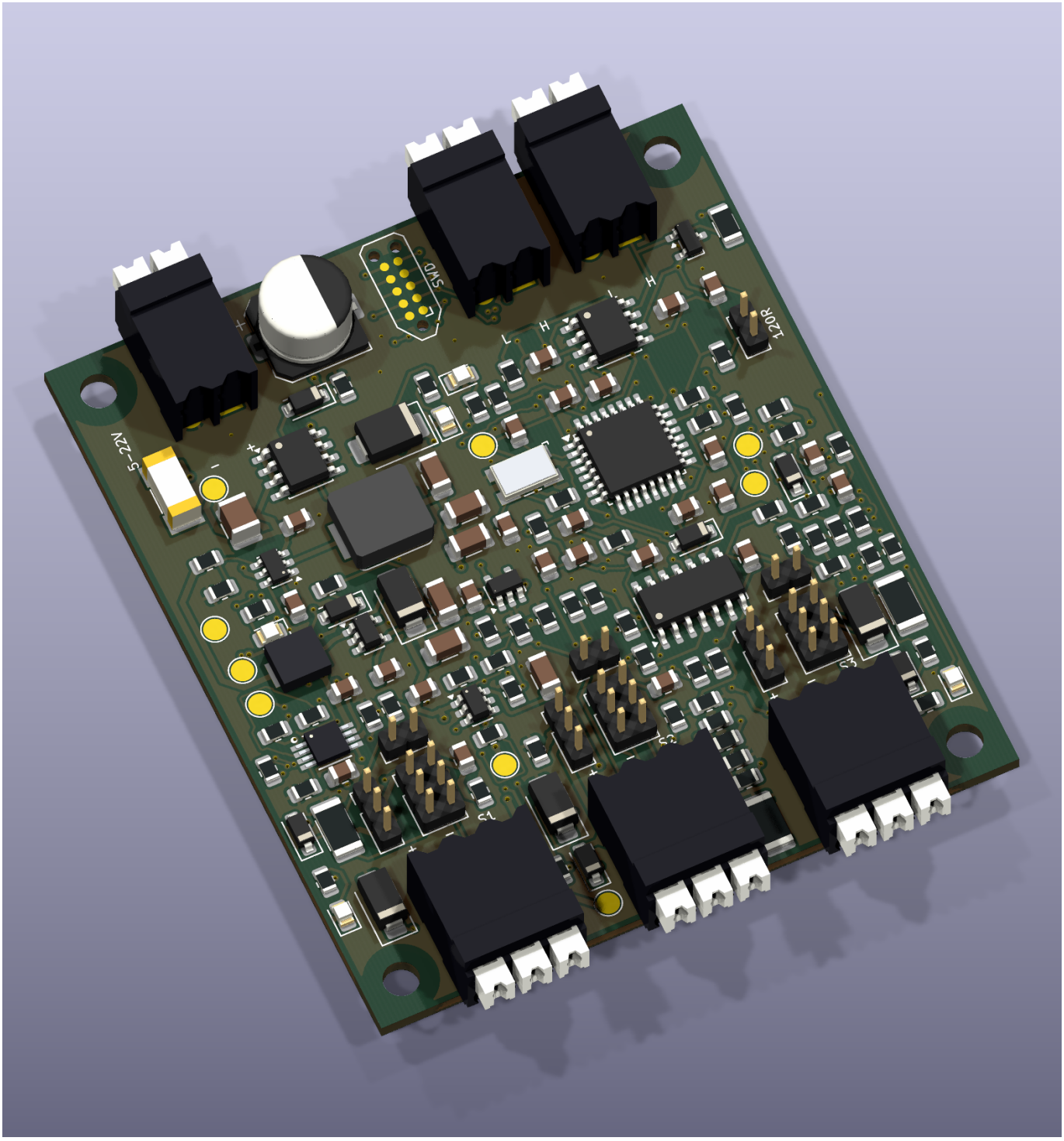
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1 Product overview

Source: README.md

Scope: Rev B. Revision overview · All revisions.

A compact three-sensor CAN interface with push-in wiring and selectable sensor power. Rev B is a 60 × 68 mm, two-layer redesign, 55.7% smaller in area than the previous 96 × 96 mm board. It is an unbuilt prototype with computational checks and manufacturing files; hardware qualification remains pending.



Feature	Rev B
Sensor channels	Three independent digital inputs

Feature	Rev B
Board input target	5–22 V DC at the connector, including tolerance and cable drop
Signal range	Positive 0–30 V, with selectable industrial or 3.3/5 V logic thresholds
Sensor power	Per-port physical selection: regulated nominal 5 V or switched VBAT
Software power control	Enable/disable and monitor each shared rail; voltage selection uses shunts
Current targets	300 mA total on 5 V and 300 mA total on VBAT, shared across the ports
External connectors	Weidmuller 3.5 mm horizontal /90 push-in, through-hole
Programming	TC2050-IDC-NL footprint, 1.27 mm contact pitch, three guide holes; custom electrical adapter
Controller / CAN	STM32F042K6T6, separate TCAN3413, original 1 Mbit/s Team Use protocol
Assembly	Mostly 0805 or larger passives and leaded ICs; heatbed/reflow required for U3's exposed pad
Parts estimate	$59.30 at observed single-unit prices, including ten shunts; excludes PCB, tools and shipping

VBAT follows the input; it is **not regulated 12 V**. Adding regulated 12 V or software voltage multiplexing would increase cost and complexity. Industrial sensors with a 10 V minimum can stop operating during battery sag even when this board remains powered. A nominal 24 V power supply is outside the board's input target. A separately powered sensor can still provide a positive signal up to 30 V with common ground and the correct configuration.

Logic-mode targets are LOW ≤0.8 V and HIGH ≥2.4 V; industrial-mode targets are LOW ≤4 V and HIGH ≥7 V. This is a digital interface; it does not promise 1.8 V logic, analog acquisition, IO-Link, quadrature or CTRE Phoenix/Tuner compatibility.

Review and build files

• KiCad project, schematic, routed PCB

• Printable handbook, BOM PDF

• Grouped BOM, DigiKey import, purchasing notes

• Interactive assembly BOM, assembly drawing, schematic PDF

• Gerbers and drills, STEP assembly

• Wiring, shunts and pogo pin map, hand assembly, firmware, robot examples

• Validation evidence, engineering decisions, bench worksheet, design plan

Use Rev B hardware, BOM and firmware together. Rev A remains available in the repository root and its archived build package. Open the supplied native board as the authoritative routed design; placement/router scripts are engineering history, not a substitute for the saved board. Rebuilding the full routing from the initial placement is not part of release reproduction.

Validation and limits

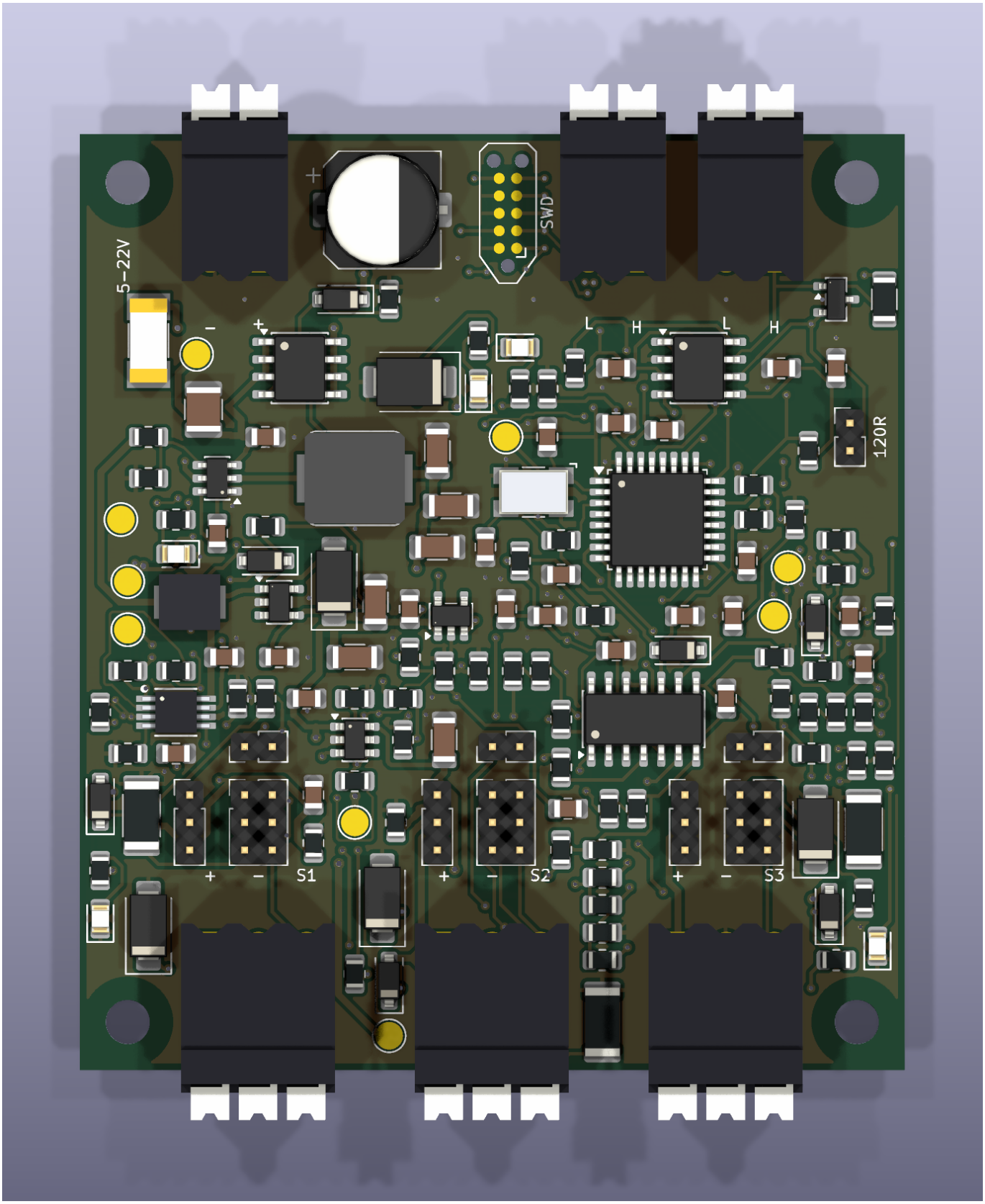
The final native ERC/DRC, model/schematic/PCB parity and independent exported-copper checks are recorded in the validation report. Firmware tests cover power faults, startup, watchdog handling, protocol behavior and the reserved fourth channel. Java, C++ and Python integration tests use simulated transports; actual CAN hardware and pogo programming have not been exercised.

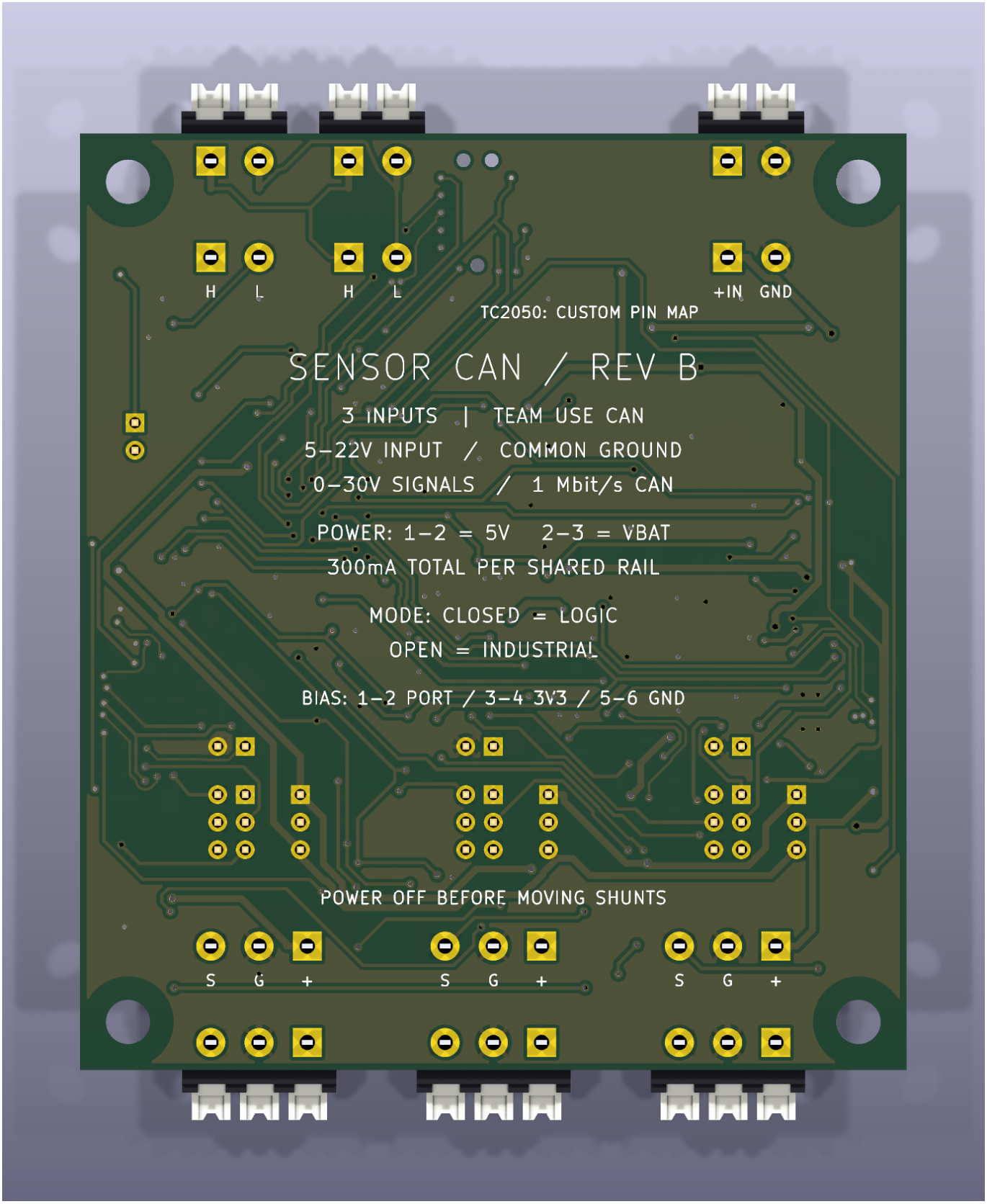
Electrical calculations support the stated targets under their recorded assumptions. The 5 V output is nominal: the full-load routed-copper budget estimates roughly 4.75 V at the most distant port, before external cable loss. Boost startup simulations settle near 4.99 V, but do not validate the cascaded converters, real parasitics or temperature behavior. Complete the bench worksheet before treating any target as a qualified product

rating.

See TOOLS.md for reproduction commands and source notices for provenance. No fabrication order or purchase has been made.

Additional board views





Connector housings use official manufacturer STEP geometry; see connector verification. Render colors are illustrative.

2 Design decisions

Source: docs/ENGINEERING.md

Scope: Rev B. Revision overview · All revisions.

The 60 × 68 mm outline occupies 4,080 mm², versus 9,216 mm² for Rev A: 55.7% less area. Three sensor ports preserve useful capacity while leaving space for push-in terminals, consistent selector placement and an accessible programming cable. Four 3.2 mm mounting holes are centered at (3.5,3.5), (56.5,3.5), (3.5,64.5), and (56.5,64.5) mm. Use insulated mounting hardware and confirm the selected washer/enclosure clearances. The terminal housings extend approximately 8.5 mm above the PCB; the 3.5 mm leads project below the top seating plane.

The design retains a STM32F042K6T6 in LQFP32 and a separate TCAN3413 CAN transceiver. An ESP32-C3 has a CAN/TWAI controller, not an integrated physical CAN transceiver. Retaining the small bare-metal STM32 implementation avoids introducing a radio/module, new firmware dependencies and another untested CAN implementation during the mechanical redesign. This is an original Team Use protocol, not CTRE compatibility.

The input path uses a fuse, reverse-polarity MOSFET and 22 V standoff TVS. The input bulk capacitor is now 50 V rated. The TVS's specified pulse clamp is 35.5 V; actual overshoot, pulse energy and wiring inductance need measurement. It is not a regulator for sustained overvoltage. The 5–22 V target is measured at the board connector, including source tolerance and cable drop. A nominal 24 V supply is unsuitable.

A LMR51420 buck produces approximately 3.97 V, then a TPS613222A boost produces 5 V. This supports input below the voltage required by a simple 5 V buck. Both converters use small leaded SOT packages, with external inductors and an SMA Schottky rectifier. The LM339LV input comparator runs from regulated 3.3 V; the sensor's 0–30 V signal is divided before reaching it. Do not substitute a 30 V signal for a 30 V comparator supply rating.

Per-port voltage selection is physical. Software can enable/disable the two rails, report their voltages, and latch faults. Safe software selection between 5 V and a 22 V rail would need additional reverse-blocking switches and interlocks on every port. The shunt solution reduces parts, heat and assembly complexity. A separate regulated 12 V output would add another converter, so this revision offers regulated 5 V or switched battery voltage.

The software protections supplement hardware current limiting. Sensor voltage health uses ADC measurements with startup blanking. The battery switch's open-load diagnostic may assert at light load even when voltage is healthy; firmware uses voltage to distinguish that condition. Fault clear is an explicit recovery action. Persistent settings can automatically enable requested rails after a normal reboot, while watchdog/fault handling forces them off as described in the firmware documentation.

The design uses 0805 or larger passives, SOIC logic/transceiver packages, LQFP32 MCU, through-hole terminals and 2 mm selector headers. U3 has an exposed pad that needs paste and heatbed/reflow; soldering only its visible leads is insufficient. Through vias close to IC lead escapes require reliable solder mask/tenting and visual inspection. The preferred prototype process is a 0.10 mm stencil, controlled paste/reflow appropriate to the actual alloy, followed by hand soldering of terminals and headers.

The deterministic checks publish their assumptions in `reports/electrical/calculations.json`. They budget both sensor rails at 300 mA, additional bias current, a generous 150 mA logic allocation, 80% efficiency per converter, and hot input-path resistance allowances. These are design estimates, not measured temperature or output-current guarantees. Comparator bias/leakage and boost ripple bounds include engineering allowances where the datasheets provide only typical values.

The TI boost model was exercised in ngspice with PSpice compatibility, `SS=0` for startup, and a syntax-only initial-condition adaptation. The original vendor file is retained. Both completed startup simulations settle near 4.99 V: approximately 420 mA with a 3.97 V source and 462 mA with a 3.80 V source. The latter has about 3.1 mV simulated steady-state ripple and 0.753 A peak inductor current. The upstream buck is represented by a finite-resistance source; the model does not establish cascaded-loop stability, temperature behavior, quiescent current or actual PCB parasitics. The sampled waveform, plot, model adaptation and measurements are included; these are model results, not hardware measurements.

Prototype qualification must cover startup at 5 V and 22 V, simultaneous rail loading, sensor inrush, brownout recovery, short circuits, rail backfeed, CAN error recovery, comparator thresholds/leakage over temperature, crystal startup/drive, power ripple, converter temperatures, wire retention and pogo alignment. Native ERC/DRC and Gerber connectivity checks cannot replace these measurements.

The routed-copper audit estimates 4.749–4.760 V at a port carrying the full 300 mA 5 V load, using the converter minimum, hot switch resistance and additional connector/shunt/ground allowance. This is a calculated budget, not a measured output guarantee. Input trace losses prompted increasing the total trace/connector/return allowance to 110 mΩ in the low-input calculation. The minimum practical sensor voltage and simultaneous-load rating still require bench verification.

The ground-return review uses two DC resistor-sheet meshes of the exported Gerbers and correlated component tolerance corners. A dedicated return trace and three additional stitches improve the sensor ground return and preserve the logic thresholds under 108 combined-load/supply cases. These calculations include a 15% ground-potential inflation allowance; they do not model switching-frequency ground bounce. The final programming escape uses straight outward contact traces, a top pour exclusion and a native 0.508 mm contact-to-foreign-track rule, checked against the TC2050 drawing.

3 Wiring and selectors

Source: docs/WIRING.md

Scope: Rev B. Revision overview · All revisions.

This revision has three independent digital sensor inputs, two shared switched power rails, and a custom 1 Mbit/s CAN interface. It is a prototype design; physical bring-up and qualification remain outstanding. It does not implement CTRE Phoenix or Tuner.

Connector	Pin 1	Pin 2	Pin 3
J1 input	+5–22 V DC	Ground	—
J2, J3 CAN	CAN H	CAN L	—
J4 sensor 1	Selected power	Ground	Signal
J5 sensor 2	Selected power	Ground	Signal
J6 sensor 3	Selected power	Ground	Signal

Use the numbered pads and board markings. J1–J3 face the opposite board edge from J4–J6, so pin numbers are more reliable than assuming the same left-to-right order. The two solder legs for each terminal pole are the same electrical pin. CAN terminals do not provide a ground pin: power grounds establish the shared reference.

Supply 5–22 V **at J1 under load**. The maximum is 22 V including supply tolerance. A nominal 24 V industrial supply is outside this input rating, although a separately powered sensor may present a positive signal up to 30 V. Use an appropriate fused FRC distribution output and common ground; the onboard 1.5 A fuse does not replace robot distribution protection.

Weidmuller /90 terminals take wires horizontally, parallel to the PCB, with push-in actuators. Sensor terminals are 1824430000; power/CAN terminals are 1824420000. Strip 8 mm. Follow the manufacturer's conductor/ferrule limits; 20–22 AWG flexible robot wiring is a practical starting point. Press the actuator to insert or remove fine-stranded wire when required. Secure cables against robot vibration and pulling. Assemble the through-hole terminals after the surface-mount parts.

Select sensor power

Each port gets its own three-pin selector. **Turn board power off before changing shunts. Fit exactly one power shunt per powered port.**

Port	Selector	Bridge pins 1–2	Bridge pins 2–3	No shunt
1	JP101	Regulated 5 V	Protected switched VBAT	Power pin disconnected
2	JP201	Regulated 5 V	Protected switched VBAT	Power pin disconnected
3	JP301	Regulated 5 V	Protected switched VBAT	Power pin disconnected

VBAT follows the input voltage, less protection and wiring losses. It is not regulated 12 V. A sensor requiring 10 V minimum can stop operating when the FRC battery sags, even though the board itself continues running. The 5 V rail has a 300 mA design target shared across all ports selecting it; the VBAT rail has a separate 300 mA shared target. These are total rail budgets, not 300 mA per port. Account for sensor inrush and bias currents.

Software switches the two shared rails on/off and reports their voltage and fault state. It does **not** change a port's voltage selection and cannot read the shunt position. Rail mask: 0=off, 1=5 V, 2=VBAT, 3=both. An erased/unconfigured board starts with both rails off. A stored nonzero rail setting can restore power automatically after ordinary startup.

Do not connect an external power source to a port power pin. For externally powered sensors, remove that port's power shunt, connect common ground and signal, and choose an appropriate bias. The battery switch and reverse-polarity MOSFET do not provide general reverse-current isolation.

Select input threshold and bias

Port	Threshold jumper	Bias header
1	JP102	JP103
2	JP202	JP203
3	JP302	JP303

Threshold jumper fitted selects logic mode: target LOW ≤0.8 V and HIGH ≥2.4 V. Jumper removed selects industrial mode: target LOW ≤4 V and HIGH ≥7 V. Voltages between the LOW and HIGH limits have hysteretic/unspecified state. Positive signal range is 0–30 V. This is digital detection, not an analog measurement input; 1.8 V logic is not guaranteed.

Fit at most one bias shunt, horizontally across one row of the 2×3 header:

Bias row	Bridge	Use
Top	1–2	Pull up through 4.7 kΩ to the port's selected supply; usually industrial NPN/open collector
Middle	3–4	Pull up through 4.7 kΩ to 3.3 V; suitable for low-leakage logic open drain
Bottom	5–6	Pull down through 4.7 kΩ; usually PNP sourcing outputs
None	Open	Push-pull output or external bias

For a 10–30 V NPN photoelectric sensor powered by VBAT: select VBAT power, industrial threshold, and top-row bias. For PNP: select VBAT, industrial threshold, and bottom-row bias. For a 5 V push-pull sensor: select 5 V power, logic threshold, and no bias. Check the actual sensor's output type, minimum supply and off-state leakage. Use the software inversion mask if the desired logical meaning is opposite to the electrical level.

Input state is sampled at 1 ms intervals. Pulses shorter than a sampling interval can be missed, and debounce adds delay. The counters follow the firmware sampling/filtering path; they are not hardware edge-capture counters. Choose sensor timing and filtering accordingly.

CAN and service connection

J2/J3 are electrically parallel for daisy chaining. Use twisted CAN H/L wiring and a short local stub. JP2 adds 120 Ω termination; fit it only at a bus endpoint. With power off, a correctly terminated whole bus usually measures about 60 Ω between H and L.

J8 uses the **standard Tag-Connect TC2050-IDC-NL mechanical footprint with a custom electrical pinout**. A generic 1:1 TC2050 cable needs an adapter wired to this table. It is not the standard ARM Cortex 10-pin mapping.

J8/cable pin	Board signal	Connect to
1	NRST	Probe reset, optional

J8/cable pin	Board signal	Connect to
2	UART_RX	3.3 V UART adapter TX
3	UART_TX	3.3 V UART adapter RX
4	SWCLK	SWD probe clock
5	SWDIO	SWD probe data
6	+3V3	Probe VTref **sense input only**
7	Ground	Probe ground
8	Ground	Service ground
9	BOOT0	Leave open normally; use only for deliberate bootloader service
10	Ground	Service ground

Power the board at J1 during programming. Leave programmer/UART power outputs disconnected. Align the cable using its three guide pins; the contacts have 1.27 mm pitch. UART is 115200 baud, 8N1, 3.3 V logic. Secure the cable with a suitable TC2050 retaining clip or fixture; a bare pogo pattern is not a permanent connector.

4 Purchasing

Source: docs/SOURCING.md

Scope: Rev B. Revision overview · All revisions.

The grouped BOM and DigiKey import CSV specify exact manufacturer and distributor part numbers. The estimate is **$59.30 per board in components**, using conservative single-unit prices, including ten 2 mm configuration shunts. PCB fabrication, stencil, tax, freight, tools, programming cable and assembly are excluded. This is a dated engineering estimate, not a reserved-stock quote.

Prices and inventory were observed on 2026-09-07/08. Regional DigiKey listings can expose different quantities, currencies and pack sizes. The selected parts favor established manufacturers and common leaded packages; global availability is a sourcing preference, not a guarantee. Recheck exact MPN, currency, cut-tape quantity, lifecycle and delivery location before placing an order.

The six horizontal push-in terminals total $11.25 at the observed quantity-one prices. Selected parts are Weidmuller LSF-SMT 3.50/02/90 3.5SN BK TU (1824420000, $1.54, 12,805 observed) and LSF-SMT 3.50/03/90 3.5SN BK TU (1824430000, $2.21, 1,698 observed). US DigiKey pages were checked on 2026-09-08. An older cached result showed the two-pole part out of stock; the freshly opened product page supplied the later observation. These are dated observations, not reserved stock. Each pole has two solder legs, 7 mm apart; this is not the previous /135 footprint.

A physically similar push-in connector is not automatically a substitute. Check entry direction, body envelope, pin rows, hole pitch, lead length, current rating and actuator clearance. Screw terminals are not included. The chosen terminals accept bare wire; use the manufacturer's 8 mm strip length and wire preparation instructions. For this prototype, 20–22 AWG is a practical assembly choice within the selected manufacturer's range.

For passive substitutions, preserve value, tolerance, temperature coefficient, power/voltage rating, dielectric and footprint. Retain the 0.1% divider resistors, higher-power 2010 pull-down resistors, 50 V input capacitor and specified effective output capacitance. MLCC nameplate capacitance alone does not establish capacitance at DC bias.

Do not substitute a comparator by pinout alone. Rev B supplies LM339LV from 3.3 V and relies on its input range. Power switches, converter suffixes, inductors and crystal/load capacitors require a new electrical review when substituted. TPS2553DBVR-1 is the latched variant.

External accessories are separate: a TC2050-IDC-NL cable, suitable SWD probe and custom adapter matching the documented pin map. The cable's mechanical fit does not imply a standard ARM electrical pinout. No US cable price is included because only a regional price was reliably observed. Insulated mounting hardware and robot-side fused power wiring are also separate.

Sources: two-pole manufacturer drawing, three-pole manufacturer drawing, three-pole DigiKey listing, Tag-Connect cable. Per-part links and observation dates are in the BOM.

5 Hand assembly

Source: docs/ASSEMBLY.md

Scope: Rev B. Revision overview · All revisions.

Use the exact Rev B BOM and board files together. Rev A has a different outline, connectors, comparator supply and power conversion. Do not mix the revisions.

1. Check the bare board, paste openings and drill sizes against the assembly drawing. Confirm that the terminal footprints have two plated holes per pole and that J8 has three nonplated alignment holes.

2. Fit the surface-mount parts with paste and controlled heatbed/reflow. Verify polarity for the input capacitor, diodes, LEDs and MOSFET. Confirm pin 1 on U3, U5 and U7. Inspect the exposed pad and thermal vias under U3, then inspect every MCU pin for bridges.

3. Fit the through-hole push-in terminals and 2 mm headers. JP302 pin 2 connects directly to the ground planes; preheat and sufficient iron power help wet this joint. Keep actuator and wire-entry directions consistent with the drawing. Clean and inspect joints before installing any shunts.

4. Leave power selectors and termination open for initial inspection. Check input-to-ground and 3.3 V-to-ground resistance for a hard short. Check that the two legs of each terminal pole are connected and adjacent poles are isolated.

5. Use a current-limited bench supply at J1, initially 12 V with no sensors. Verify protected input, approximately 3.97 V intermediate rail, 5 V internal rail and 3.3 V logic. With erased firmware/configuration, sensor power outputs should remain off.

6. Attach the correctly wired TC2050 adapter and SWD probe. Confirm target voltage/device identity, program the Rev B image, and read back/verify flash. Keep probe power disconnected. Provision a unique CAN device ID with rails=0.

7. Power down, configure one sensor port, then reconnect power. Explicitly request the required rail. Verify voltage at the terminal before connecting the sensor. Test NPN/PNP or logic behavior against the selected threshold and bias.

8. Repeat for all three ports, then test combined loads at 5, 6.3, 12, 14.4, 20 and 22 V input. Confirm current limit, fault latch and explicit recovery with controlled loads. Measure temperatures and ripple rather than relying on touch or LED state.

9. Connect CAN with proper bus termination, verify all state bits/counters and diagnostics, and test communication loss, Broadcast Disable, watchdog reset and brownout recovery before robot use.

Do not change power or bias shunts while powered. Do not use a programmer's 3.3 V power output to feed the board. The delivered design has not been physically assembled; record measured results in the bench worksheet before considering it qualified.

Horizontal connector clearance: PCB remains 60 x 68 mm. Reserve an assembled envelope of 60 x 76.1 mm before adding cable bend/service space. The fixed housings project about 1 mm beyond each connector edge; the conservative actuator envelope extends 4.05 mm. Seated connector height is 8.5 mm, with 3.5 mm leads before trimming. Do not clamp or obstruct the front actuators.

6 Firmware and programming

Source: firmware/README.md

Scope: Rev B. Revision overview · All revisions.

Bare-metal STM32F042K6 firmware for three inputs. The fourth protocol channel is reserved and always zero. Two final 1 KiB flash pages hold redundant CRC-protected configuration. The image uses 7,148 bytes of flash and 1,832 bytes of RAM including reserved stack; these are linker allocations, not measured stack depth.

From `rev_b/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

The build produces `firmware/build/sensor_can.elf`, `.hex`, and `.bin`. Host tests simulate peripheral boundaries; they do not qualify real STM32 registers or wiring.

Power J1 from a current-limited bench supply. Use the custom TC2050 adapter in WIRING.md: probe VTref sense→J8.6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected. The mechanical footprint is standard TC2050-IDC-NL; the electrical mapping is custom.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -c 'program firmware/build/sensor_can.elf verify reset exit'
```

This has not been exercised on physical Rev B hardware. Verify target identity, voltage and flash readback. Normal application updates preserve configuration; a full-chip erase removes it.

For UART, connect a 3.3 V adapter's TX→J8.2, RX→J8.3, ground→J8.8 or 10. Leave adapter power disconnected. UART is 115200 baud, 8N1.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2 --rails 0
```

Device IDs are 0–63 and must be unique for this Team Use protocol. Rev B inversion masks are 0–7; bit 0 is sensor 1. Debounce is 1–100 ms. Rail masks: 0=off, 1=5 V, 2=VBAT, 3=both. Voltage selection is physical. UART `SET id invert debounce_ms rails` disables rails, writes configuration, and resets. A stored nonzero rail mask enables those rails after ordinary startup; provision rails=0 until wiring is verified. Avoid repeated persistent writes from robot code.

Unconfigured boards keep both rails off and CAN in standby. Valid configuration enables the original 1 Mbit/s, 29-bit Team Use protocol. Inputs are sampled at 1 kHz; counters report debounced logical rising edges. Input LEDs show logical state. This is not an encoder, IO-Link or analog-input device.

Hardware pulldowns establish off defaults. Clock failure, hard fault, PVD low voltage and a foreground stall longer than 20 ms force rails off; the independent watchdog provides reset recovery. Enabled rails below a measured 4.5 V latch off after startup blanking. Battery light-load FAULT is distinguished using output voltage. Broadcast Disable turns both rails off and clears the requested rail state. Recovery needs an explicit rail request and fault clear after the off-time requirement. Do not continuously retry faults.

CAN initialization failure can leave deliberately configured sensor power requested while communication is unavailable. Robot code must check fresh status, diagnostics and the sensor's supply requirements. Reading data never enables power. See CAN_PROTOCOL.md for packets and recovery policy.

7 CAN protocol

Source: docs/CAN_PROTOCOL.md

Scope: Rev B. Revision overview · All revisions.

Rev B has three physical channels. Only inversion/state bits 0–2 are used. The fourth state bit and fourth counter slot stay zero for protocol compatibility. Firmware identity is 0.2.0, hardware 2.

FRC Sensor CAN protocol version 1

The interface uses 1 Mbit/s Classical CAN with 29-bit extended data frames. It is an original Team Use protocol, with device type 10 (Miscellaneous) and manufacturer 8 (Team Use), following the WPILib addressing layout. It does not implement CTRE Phoenix or impersonate CTRE hardware. Coordinate IDs with any other team devices using this same type/manufacturer pair.

```
identifier = (10 << 24) | (8 << 16) | (api << 6) | device_id
```

`device_id` is 0–63, `api` is 0–1023. For example, device 17 status is `0x0A084011`. All team packets below have DLC 8. Multibyte integers are unsigned little-endian. Reserved bytes/bits must be zero. Invalid frame type, length, version, command magic, ranges or reserved bits are rejected. Receive filters reserve a separate FIFO for the standard FRC Broadcast Disable message.

API	Direction	Period	Bytes 0 through 7
0x100 status	Board → robot	10 ms	protocol=1, sequence, logical inputs, raw inputs, enabled rails, latched faults, health flags, reset flags
0x101 counts 1/2	Board → robot	100 ms	channel 1 uint32, channel 2 uint32
0x102 counts 3/4	Board → robot	100 ms	channel 3 uint32, reserved zero uint32
0x110 diagnostics	Board → robot	100 ms	switched battery output mV uint16, sensor 5 V mV uint16, TX queue drops uint16, TEC uint8, REC uint8
0x111 identity	Board → robot	1000 ms	protocol=1, firmware major=0, minor=2, patch=0, hardware=2, device ID, debounce ms, inversion mask
0x200 runtime configuration	Robot → board	On explicit change	0x53, 0x43, 1, inversion mask, debounce low byte, debounce high byte, requested rails, 0
0x201 clear fault	Robot → board	On explicit request	0x53, 0x43, 1, clear mask, 0, 0, 0, 0

Input bits 0–2 mean channels 1–3; bit 3 is reserved zero. Raw inputs are comparator levels before inversion and debounce. Logical inputs are the stable, debounced states after XOR with the configured inversion mask. Rising-edge counters count changes to logical TRUE; they start at zero on boot and wrap modulo 2^{32}. Counter-pair frames are separate snapshots, not an atomic snapshot of all channels. Runtime configuration reinitializes debounce state while preserving counts. Input LEDs show logical state.

Rail bit 0 means regulated 5 V; bit 1 means battery. An enabled bit reports the MCU's commanded switch-enable state, **not measured power-good**. A hardware switch can have shut down before software updates the status. Latched faults require manual recovery. The battery diagnostic measures SENSOR_VBAT after the battery switch. The 5 V diagnostic measures the shared switched 5 V rail. Both are measured on the PCB; cable/connector loss and a disconnected sensor remain outside that measurement. Firmware samples

both every 1 ms and latches an enabled rail off below a measured 4.5 V after its 5 ms startup blanking interval. ADC failure latches both rails off. The battery switch also asserts its digital FAULT for healthy loads below 10–20 mA, so battery fault handling uses output voltage while the switch retains autonomous current-limit/latch protection.

Health flags: bit 0 configured, bit 1 clock healthy, bit 2 ADC healthy. Reset flags: bit 0 power-on, bit 1 reset pin, bit 2 independent watchdog, bit 3 software reset; several bits can be set. Diagnostic ADC readings are approximate monitoring values, not calibrated instrumentation. Ignore voltage readings if ADC health is false. TX queue drops count local mailbox submission failures, saturating at 65535; they do not count every unacknowledged or corrupted transmission. TEC and REC are controller error counters.

Debounce is 1–100 ms with 1 kHz sampling. The default service example uses 2 ms. Analog filtering and sample phase add latency; seven typical-model SPICE cases observed roughly 2–3.1 ms transition latency with this setting. No pulse-width, quadrature, encoder speed or minimum-pulse guarantee is established. The status sequence increments modulo 256 and is a freshness aid, not a transaction identifier.

Runtime configuration accepts inversion 0–7 and rails 0–3. It is volatile and does not clear a latched rail. To recover a fault: remove the cause, leave the rail off for at least 100 ms, explicitly request the desired rail state, then explicitly clear its fault bit. Clear masks are 1–3. Do not retry automatically in robotPeriodic. UART SET is the only normal persistent configuration operation and resets the board after a verified flash write.

Timing and short pulses

Status is a periodic sample, so a brief debounced pulse can occur between 10 ms transmissions. Rising-edge counters retain such events until reset but arrive every 100 ms and have no event timestamps. Local debounce latency also excludes CAN and robot-loop delays. See the sampling and timing notes before choosing a sensor geometry or event-handling strategy.

Broadcast Disable and startup

Extended data identifier **0x00000000**, with any DLC 0–8, is FRC Broadcast Disable. Firmware immediately commands both rails off, sets both latches and clears requested rails. Communication remains available. Recovery requires an explicit rail request and fault clear after the off-time requirement. Ordinary disabled-state heartbeat does not switch sensor power in this sensor-only design. This device has no actuator outputs.

With no valid flash configuration, CAN remains in standby and sensor rails remain off. Assign a unique ID through UART before connecting several unconfigured boards to a robot. A watchdog reset keeps both sensor rails latched off even if stored configuration requests them. Clock or low-voltage faults force safe shutdown. Automatic CAN bus-off recovery is enabled, but sensor data must still be treated as unavailable when stale.

Robot-side rules

Use a 100 ms status freshness limit; this is implemented in the supplied Java and C++ wrappers. Counter/diagnostic freshness is 350 ms and identity freshness is 1500 ms. Missing or invalid packets return no sample. Do not substitute a stale FALSE for a sensor failure. Verify configured, clock and ADC flags, the relevant rail enable/fault bits, and adequate sensor supply before acting. The supplied intake examples fail closed when their assumed 10 V sensor supply margin is absent.

There is no command acknowledgment packet, transaction ID, encryption or authentication. Read status and identity back after configuration. This is intended for a trusted, physically controlled robot CAN bus. Periodic traffic is 131 frames/s; a conservative 160 bits/frame estimate gives about 2.1% bus load per board. Team-controlled command rates should remain low.

Sampling and event timing

Scope: Rev B. Revision overview · All revisions.

Inputs are sampled every 1 ms. Firmware applies configurable 1–100 ms debounce, sends status every 10 ms and sends each pair of rising-edge counters every 100 ms. The service example selects 2 ms debounce. There are three active inputs; the fourth protocol slot remains zero.

A short pulse can fall between samples. A pulse detected by one sample can also disappear before the debounce interval completes. Allow at least the configured debounce interval plus a sampling interval, analog/sensor delay and margin; establish the actual minimum pulse width on hardware. A 5 ms pulse is a useful initial bench test with 2 ms debounce, followed by shorter and longer pulses at varied phase. Rev B has no measured pulse-capture guarantee.

A debounced pulse can occur entirely between two status packets. Rising-edge counters retain a captured logical rising event until reset, but their slower reporting adds delay and they carry no timestamp. Select inversion so the event of interest is a rising edge. Establish a counter baseline on startup and after data loss, handle unsigned rollover, and reject implausible jumps after a board reboot. The two counter-pair messages are separate samples.

With 2 ms debounce, local firmware response is roughly 2–3 ms after the sampled electrical transition, excluding analog delay. The next status transmission can add approximately 10 ms and a 20 ms robot loop can add almost another 20 ms. CAN arbitration and transport delay add to this scheduling budget; it is not a measured or guaranteed end-to-end bound.

Check the shortest beam interruption at the highest expected game-piece speed. Test the complete sensor, wiring, board, CAN and robot loop. These filtered digital inputs and telemetry rates are not qualified for PWM, quadrature or high-speed encoder acquisition. The Rev A comparator pulse study is not claimed as Rev B validation.

8 Robot integration

Source: examples/README.md

Scope: Rev B. Revision overview · All revisions.

Robot integration — WPILib 2026.2.1

The Java and C++ interfaces read this board's original Team Use CAN protocol. They compile against WPILib 2026.2.1 APIs. Host tests replace the transport boundary; an actual roboRIO integration test remains required.

For Java, copy `java/src/main/java/frc/robot/sensors/SensorCan.java` and `SensorCanCodec.java` into the matching package in a Java 17 robot project. Construct one `SensorCan` per physical board as a subsystem member, using the ID provisioned through UART. `status()`, `diagnostic()`, `counts12()`, `counts34()` and `identity()` return `Optional<Sample<...>>`. Missing, malformed or stale data returns empty. Close the object when disposing the subsystem.

For C++, copy the two headers from `cpp/include` into the robot project and construct `sensorcan::SensorCan` as a subsystem member. `ReadStatus()`, `ReadDiagnostic()`, `ReadCounts12()`, `ReadCounts34()` and `ReadIdentity()` return `std::optional<Sample<...>>`. The class uses the project's normal WPILib CAN library; no additional runtime library is required.

The intake examples in both languages assume channel 1 is logically TRUE when a game piece is present, with a battery-powered sensor requiring at least 10 V. They permit intake only when fresh status and diagnostics exist, configured/clock/ADC flags are healthy, battery power is enabled without a fault, switched battery output reports at least 11 V, and the piece is absent. The 11 V check is an example allowance for switch/wiring drop and diagnostic error; measure your actual sensor and adjust its policy deliberately. Neither example is a complete motor-control subsystem.

Reading data never enables sensor power or clears faults. `configureRuntime` / `ConfigureRuntime` sends one volatile configuration request; `clearFaults` / `ClearFaults` sends one explicit recovery attempt. Do not call either continuously from periodic code. Verify the resulting status/identity, and expose failures to the driver station. A constructor or successful packet write does not prove the board is present.

Status expires after 100 ms; counter/diagnostic packets after 350 ms; identity after 1500 ms. Counters are unsigned 32-bit totals and wrap. Java represents them as `long`; C++ uses `uint32_t`. Counter pairs arrive separately. Do not use these 1 kHz debounced inputs as high-speed encoders or rely on them as a safety interlock.

The Python service tool can also passively monitor a Linux SocketCAN interface already configured for 1 Mbit/s:

```
python host/sensor_can.py monitor --channel can0 --device 17 --seconds 30
```

Configure the CAN adapter using its own supported procedure. The monitor sends no commands and reports unavailable status after 100 ms without a valid frame. UART provisioning is described in firmware/README.md.

Run `bash scripts/test_drivers.sh` from the project root to compile and exercise Java, C++ and Python integrations using the installed toolchain. No LabVIEW VI is supplied; a LabVIEW implementation would need to implement the documented packets, freshness checks and fault policy, then be tested on the robot. This project does not claim complete ecosystem certification.

9 Qualification worksheet

Source: docs/QUALIFICATION.md

Scope: Rev B. Revision overview · All revisions.

Status: **not physically built or tested**. Record board serial number, PCB lot, firmware hash, instruments, ambient temperature and operator for every run. Start with current-limited laboratory supplies and controlled loads. Exported design checks establish file consistency; they do not establish a production rating.

Test	Method and acceptance target	Measured result
Assembly	Inspect polarity, MCU bridges, exposed-pad wetting, fuse, terminal double legs, solder-mask coverage and all seven NPTH holes. Confirm no shorts before applying power.	Pending
No-load supplies	Sweep J1 through 5, 6.3, 12, 14.4, 20 and 22 V. Record input current, intermediate rail, internal 5 V and 3.3 V; no oscillation or abnormal heating.	Pending
Full shared-rail loads	Apply 300 mA to 5 V and 300 mA to VBAT simultaneously. Repeat each sensor port carrying the full allowed shared-rail load. Measure voltage at the external terminal, including wire and shunt loss.	Pending
Low-input startup	Cold-start at 5 V at all load combinations and repeat with realistic source/wire resistance. Inspect both switching nodes and ripple.	Pending
Voltage transients	Sweep and interrupt the input; capture MOSFET, TVS and converter voltages with appropriate probes. Confirm absolute maxima and recovery. The TVS rating does not establish load-dump immunity.	Pending
Short/inrush	Test each shared output with controlled shorts and representative sensor capacitance. Verify hardware limit/latch, firmware fault report and explicit clear. Evaluate the 40 µF external VBAT capacitance target at 22 V.	Pending
Thermal	Repeat full-load tests over the intended enclosure/ambient range. Record regulator, rectifier, inductor, switch, fuse and connector temperatures against datasheet limits with margin.	Pending
Input thresholds	For every channel and setting, ramp 0–30 V in both directions and record switching/hysteresis. Verify logic LOW at 0.8 V/HIGH at 2.4 V and industrial LOW at 4 V/HIGH at 7 V over supply/temperature.	Pending
Sensor bias	Exercise NPN, PNP, push-pull and open-contact examples with every valid bias setting. Measure leakage/disconnection behavior. Confirm external outputs tolerate the chosen pull-up voltage.	Pending

Test	Method and acceptance target	Measured result
Backfeed	With board off and rails disabled, apply representative externally powered sensor signals within the specified range. Measure logic and rail backfeed; no unintended powering. Never connect two power sources to a selected sensor rail.	Pending
Firmware failures	Force foreground stall, watchdog reset, PVD/brownout, crystal/CSS failure and a latched output fault. Check documented safe-off behavior and recovery.	Pending
CAN	Test on a correctly terminated 1 Mbit/s robot bus with realistic traffic. Exercise counters, filtering, loss/recovery, duplicate-ID handling and Broadcast Disable. Verify custom Team Use identifiers on a real roboRIO.	Pending
Programming	Repeatedly mate the TC2050 cable, verify orientation, target sense and custom adapter continuity, flash/read back, and inspect pogo-pad wear.	Pending
Mechanical	Confirm terminal wire retention, actuator access, mounting/enclosure clearance, shunt retention and vibration behavior. Review fabrication-house CAM before volume ordering.	Pending

Record corrections and rerun affected checks after any failure. The fixed 5 V converter's tolerance and board drops mean the external output is nominal 5 V; applications requiring a strict minimum need measured limits. VBAT follows the input and is never regulated 12 V. Current limits are shared across ports selecting the same rail.

10 Validation evidence

Source: docs/VALIDATION.md

Scope: Rev B. Revision overview · All revisions.

These results apply to the supplied Rev B native board and its exported files. They are computational evidence for an unbuilt prototype. Physical soldering, temperature, surge behavior, real sensor/CAN traffic and programming-cable mating remain untested.

Check	Result	Evidence
Native KiCad 10.0.6 ERC / DRC	No reported violations or unconnected items under the saved project rules	`reports/erc/final.json`, `reports/drc/final.json`
Model / schematic / board parity	154 modeled components, 403 model pins, 87 named functional nets; no mismatches	`reports/design_parity.json`
Fabrication and assembly parity	Nine graphic layers, 307 plated and seven nonplated drills, 144 purchased placements and 144 local model assignments	`reports/manufacturing/verification.json`
Independent copper reconstruction	No shorts, opens or unidentified copper islands	`reports/manufacturing/gerber_connectivity.json`
Independent manufacturing geometry	Copper clearance, edge/mount spacing, annular rings, hole web, mask/paste checks passed	`reports/manufacturing/gerber_geometry.json`
Programming contact geometry	Ten contacts, no paste, minimum foreign trace/via clearance 0.751 mm versus 0.508 mm requirement	`reports/programming_geometry.json`
Electrical component corners	8,192 evaluations in each input mode, power/protection assertions passed	`reports/electrical/calculations.json`
Ground-loaded input thresholds	All channels and both modes pass 108 supply/load allocations with correlated ground potentials	`reports/electrical/loaded_input_thresholds.json`
Ground sheet estimate	0.10 / 0.075 mm meshes, analytic calibration and reciprocity/residual checks	`reports/electrical/ground_return_model.json`
Boost startup model	Approximately 4.99 V at 420 mA and 462 mA modeled loads	`reports/boost_simulation.json`
Firmware	Builds for STM32F042; 2,190,989 core assertions plus 19 application fault/startup scenarios pass with sanitizers	`reports/firmware-build.log`, `reports/firmware-tests.log`
Host / robot drivers	232,928 Java assertions, 75,828 C++ assertions, five Python tests; WPILib 2026.2.1 compile checks	`reports/driver-tests.log`

Electrical results

Mode / channel	Rising threshold range (V)	Falling threshold range (V)
Logic / 1	1.569–2.261	0.937–1.447
Logic / 2	1.496–2.240	0.863–1.425
Logic / 3	1.529–2.240	0.896–1.425
Industrial / 1	5.349–6.784	4.715–5.966
Industrial / 2	5.336–6.779	4.702–5.961
Industrial / 3	5.215–6.778	4.581–5.959

The fine-grid estimate gives a worst sensor-ground-to-input-ground resistance of 20.28 mΩ at 85 °C copper temperature. The maximum route change between the last two meshes is 2.43%. Loaded-threshold analysis inflates the modeled ground potentials by 15%, tests both rail load extremes at all port allocations and uses 5, 12 and 22 V inputs. It includes a conservative 150 mA logic allocation and 80% conversion-efficiency assumptions. Switching-frequency ground bounce, external wiring drop and hardware tolerances outside the stated allowances are not represented.

The nominal 5 V output budget includes the converter minimum, actual routed positive-trace resistance, 135 mΩ switch resistance and 100 mΩ for contact/shunt/ground allowance. Its estimated minimum is approximately 4.75 V at a port carrying the whole 300 mA shared load. The input calculation uses 110 mΩ for trace/connector/return allowance in addition to the fuse and MOSFET. These are budgets requiring measurement, not production limits.

The boost runs use TI’s typical transient model and a finite-resistance upstream source. The 3.80 V / 462 mA case reports 4.990 V average, about 3.1 mV modeled steady-state ripple and 0.753 A peak inductor current. It does not establish stability of the complete cascaded converters or thermal performance. The original vendor model, syntax adaptation, decks, sampled data and plots are included.

Interpretation and configured checks

Native DRC keeps clearance, connectivity, courtyard overlap, holes, mask and silkscreen checks enabled. Informational/model-category checks disabled in the project are: missing_courtyard, track_not_centered_on_via, tuning_profile_track_geometries, footprint_filters_mismatch, footprint_type_mismatch. Native ERC disables: single_global_label, four_way_junction, simulation_model_issue, footprint_filter. The separate parity and fabrication checks complement these settings. No clearance or unconnected-net violation is waived.

Excellon output rounds coordinates and diameters to 0.001 mm. The independent audit accepts at most half that unit and records the actual rounding error. Gerbonara reports KiCad’s G90 header-placement warning; drill positions, diameters and plating types still match. Polygon arc approximation is 1 µm; reported annular-ring minima are interpreted within that numerical tolerance.

Firmware occupies 7,148 bytes of flash and 1,832 bytes of RAM including the reserved stack allocation. Host tests execute the application with controlled peripheral substitutes; they do not emulate all STM32 electrical behavior. The fourth protocol slot remains reserved and zero on this three-channel board. Actual SWD flashing, roboRIO traffic, fault currents and sustained timing must be checked on hardware.

Historical draft reports and routing experiments in the development workspace may contain failures. The delivery archive includes the final reports identified here. Complete `docs/QUALIFICATION.md` before promoting the design from prototype to a qualified product.

11 Tools and reproduction

Source: TOOLS.md

Scope: Rev B. Revision overview · All revisions.

The saved native schematic and routed PCB are authoritative. Reproduction starts with those files and regenerates reports and exports; it does not rerun the placement experiments or autorouter.

Tools used: KiCad 10.0.6 with matching Python bindings and standard libraries, Arm GNU GCC 13.2.1, GNU make, ngspice 42, Python 3.12, Java 17, native GCC/G++ with sanitizers, InteractiveHtmlBom 2.11.2 and Poppler. Python packages are pinned in `tools-requirements.lock`. Freerouting 2.4.1 and CadQuery 2.8.0 were design aids. Manufacturing verification uses Gerbonara 1.6.3 and Shapely 2.1.2; the DC ground estimate uses PyAMG 5.3.0.

`tools/env.sh` adds a local Python environment and `$CANDI_TOOLS_ROOT/bin` to PATH. In this workspace it can reuse the parent `.venv`. On another workstation, install native tools and provide wrappers named `kicad-python`, `host-gcc`, `host-g++` and `ibom` as needed. `kicad-python` must use the interpreter matching the installed KiCad bindings; do not mix its compiled extensions with Python 3.12. Native tool binaries and library caches are not bundled.

```
source tools/env.sh
python scripts/build_bom.py
python calculations/electrical_checks.py
make -C firmware
bash scripts/test_firmware.sh
bash scripts/test_drivers.sh
bash scripts/export_manufacturing.sh
kicad-python scripts/export_board_manifest.py
python scripts/verify_manufacturing.py
python scripts/verify_gerber_connectivity.py
python scripts/verify_gerber_geometry.py
kicad-python scripts/check_programming_geometry.py
kicad-python scripts/audit_power_routes.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python calculations/ground_return_model.py .10 .075
python calculations/loaded_input_thresholds.py
```

Driver dependency versions are pinned in `tools/driver_dependencies.json`; use `python tools/fetch_driver_dependencies.py` to restore missing dependencies. Firmware third-party sources and their provenance are included. Firmware source/build commands are detailed in its README; program only after inspection and with the documented custom Tag-Connect adapter.

The boost model runs from its directory so ngspice reads the supplied `.spiceinit`:

```
cd simulation
ngspice -b boost.cir
ngspice -b boost_corner.cir
```

Large transient raw files are omitted from the delivery ZIP. The model, circuit decks, sampled waveform, measurements and plots are included. Running the simulations recreates the raw files. Allow several GB of memory for simulations and the fine-grid ground calculation. Neither simulation includes physical PCB measurements.

To rebuild the PDFs after exports and review renders:

```
bash scripts/render_board.sh
python scripts/build_bom_pdf.py
python scripts/build_handbook.py
```

The documentation uses local DejaVu fonts from the standard Linux font directory. The PCB models resolve through `${KIPRJMOD}/library/3d`; they should not require this development machine's paths. Horizontal /90 connector models use manufacturer STEP geometry with a rigid translation and project colors; run `python scripts/import_connector_models.py` to recreate them. Other custom models, including inductors, remain nominal envelopes. CAD checks do not establish physical fit.

To check refreshed documentation and package the revision, run:

```
python scripts/review_documents.py
python scripts/build_release.py
```

The review checks local links in every maintained guide and PDF text bounds; visually inspect the affected pages too.

12 Capacitor review

Source: docs/PASSIVE_REVIEW.md

Scope: Rev B. Revision overview · All revisions.

This design has 34 capacitors. They are not all interchangeable filtering capacitors, and the present count is not proven to be the absolute minimum. This audit retains them until a reduction has measurement evidence.

References	Job	Decision
C1 C2 C3	Battery/input filtering	Keep
C4 C5 C13	Buck output reservoir	Keep; C13 is a bench optimization candidate
C6	Buck bootstrap	Keep
C7 C8	3.3V regulator input/output	Keep
C9 C10	Battery sensor switch bypass/output	Keep
C11 C12	5V sensor switch bypass/output	Keep
C14	Hardware fault timing	Keep
C15 C16 C17	Boost output reservoir/bypass	Keep
C20 C21 C22 C23	Microcontroller supply bypass	Keep
C24	Microcontroller local bulk	Keep; bench optimization candidate
C25 C26	Crystal load	Keep
C27	Reset filtering	Keep
C28 C29	Supply measurement filtering	Keep
C40 C41	CAN supply bypass	Keep
C42	CAN local bulk	Keep; bench optimization candidate
C50	Comparator supply bypass	Keep
C101 C201 C301	Sensor input filtering	Keep

Several 100 nF capacitors share a supply net because they sit beside different chips or supply pins. Copper traces have inductance: one distant large capacitor cannot supply every fast local current pulse. Other capacitors are part of the converter, crystal, fault timer or input filter rather than rail bypass.

C13, C24 and C42 are candidates for a measured reduction experiment. Test supply extremes, both sensor rails loaded, startup, load steps, ripple, CAN traffic, reset behavior and temperature before approving their removal. No capacitor has been removed on the basis of appearance or a DC continuity check.

Most small resistors set input thresholds/hysteresis, sensor pull-up or pull-down current, regulator output voltage, fault current limits, safe enable states, ADC scaling or LED current. Many small black parts in the render are resistors, not capacitors. A factory-assembly variant can use lower-cost stocked equivalents without deleting these functions.

The larger simplification opportunities are the input-configuration network and the two-stage 5 V supply architecture. These require a circuit redesign and validation; changing to smaller package sizes alone does not remove their functions. See capacitor_audit.csv in the BOM folder for every capacitor and its reason.

13 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

Scope: Rev B. Revision overview · All revisions.

Sources were consulted on 2026-09-07/08. Exact purchase SKUs, product-page links, stock observations and quantity-one prices are stored in `bom/catalog_selected.json` and `BOM_grouped.csv`. They are dated distributor observations, sometimes from indexed listings, not reserved inventory or proof of worldwide delivery. All fitted purchase lines use DigiKey order codes. Alternative MPNs are candidates for engineering review, not approved substitutions.

Primary technical references

Subject	Primary source	Used for
Functional reference	CTRE CANdi, user guide	Sensor-to-CAN role and regulated/battery sensor outputs; no inherited ratings/protocol
Robot electrical rules	FIRST 2026 manual	Protected power branch, wiring, isolation and CAN constraints; recheck season rules
CAN addressing	WPILib CAN device addressing	Team Use manufacturer, device type, Broadcast Disable
Robot CAN API	WPILib 2026.2.1 source, Java CAN API	Actual Java/C++ transport interfaces
MCU	STM32F042x4/x6 datasheet, ES0243 errata	LQFP32 pin map, voltage/clock/ADC limits, calibration delay
CAN PHY	TCAN3413 datasheet	SOIC8 pin map, supply, standby and bus protection
Buck regulator	LMR51420 datasheet	X=500 kHz PFM variant, L/C, reference, layout and current limits
Logic regulator	AP2112 datasheet	SOT25 pin map and output-capacitor requirements
Battery switch	TPS1H200A-Q1 Rev. E	DGN pin map/land pattern, limit, latch delay, diagnostics
5 V switch	TPS2553 datasheet	DBVR-1 latch variant, current-limit equations, fault timing
Comparator	LM339LV datasheet	3.3 V operation, failsafe input range, quad pin map
Boost	TPS61322 datasheet, TI transient model	Fixed 5 V converter, limits, startup model
Boost inductor	SRP4020TA	4.7 µH ratings and custom land dimensions
Reverse protection	AO4485	MOSFET pinout, gate/drain ratings, resistance and pulse limits
Fuse	Littelfuse 451/453 datasheet	1.5 A package, interrupt rating, resistance and nominal melting integral
Bias diode	BAT46W	100 V reverse rating, forward drop and package
CAN protection	ESD2CAN24-Q1	SOT23 H/L/GND mapping and bus protection
Inductor	SRP7028A	10 µH current ratings and mechanical outline
Crystal	ABM3B	8 MHz, 18 pF load, ESR/C0, drive level and outline
Connectors	Weidmuller 1824420000, 1824430000	Exact 3.5 mm push-in geometry, two solder legs per pole, horizontal /90 entry
Programming contact	Tag-Connect TC2050-IDC-NL	Contact pitch, alignment holes, cable envelope and copper clearance; custom electrical mapping
Capacitor derating	Samsung CL31A226KAHNNNE manufacturer characteristics; original Rev A research file retained in the development workspace	30 µF effective capacitance is an engineering simulation assumption for 44 µF nominal; confirm actual bias/temperature loss
Aluminum capacitor	Panasonic FK series	EEE-FK1H470P dimensions, ripple/impedance and handling cautions
ESP32 comparison	Espressif ESP32-C3 TWAI	CAN controller still requires an external transceiver

Local datasheets and extracted text are research references. Earlier research remains in the parent development workspace; only the current BOM and circuit determine what to purchase. The release manifest

records included source-file hashes. Changed-part PDFs are retained in `sources/`; the research file named `weidmuller_3p_90.pdf` is a rejected different-pole-count variant and is excluded from delivery. Manufacturer documents and models retain their owners' copyright and terms; they are not relicensed as original project material.

Libraries and software

KiCad standard footprints and package models came from the verified KiCad 10.0.6 distribution. Their original names are retained in each part's `source_footprint` field. Project-local copies make the design portable. The KiCad library license is CC BY-SA 4.0 with its design-output exception; the notice and legal text are retained in `licenses`. Original custom footprints cover the selected Weidmuller terminals and TI exposed-pad package. The Abracon crystal footprint adapts the KiCad body/model to the manufacturer's recommended land pattern, with its derivation recorded in the footprint provenance file. Original simplified STEP models retain their provenance; changed Rev B models are listed in `hardware/library/3d/rev_b_models.json`. They are not supplier-certified models.

Firmware includes ST CMSIS device headers/startup and Arm CMSIS headers with their original license files. WPILib headers/jars are used for integration compilation and retain the WPILib notice. InteractiveHtmlBom generates the standalone assembly page; its license is included. TI's original TPS613222A model and copyright header are preserved. The ngspice-compatible copy changes initial-condition syntax, and the test deck supplies the startup parameter documented in the engineering report. See `licenses/` and the respective third-party directories for the complete notices.

No blanket public license has been selected for the user's original hardware, firmware or documentation in this repository. Third-party licenses continue to apply to their own material. The project is versioned in Git; no physical board has been built and no manufacturing order has been placed. CTRE, FIRST, WPILib, DigiKey and component names identify compatibility targets or sources, not endorsement.

Horizontal connector CAD refresh

Official Weidmüller STEP geometry now replaces the simplified /90 terminal envelopes. The original supplier files, transformed-model hashes, public source URLs and alignment method are retained in vendor_connectors.json. Geometry is translated only; display colors are project-assigned. Supplier material retains its original ownership and is not relicensed by this repository. See CONNECTORS.md. Other custom models remain nominal project envelopes.

Connector geometry and sourcing

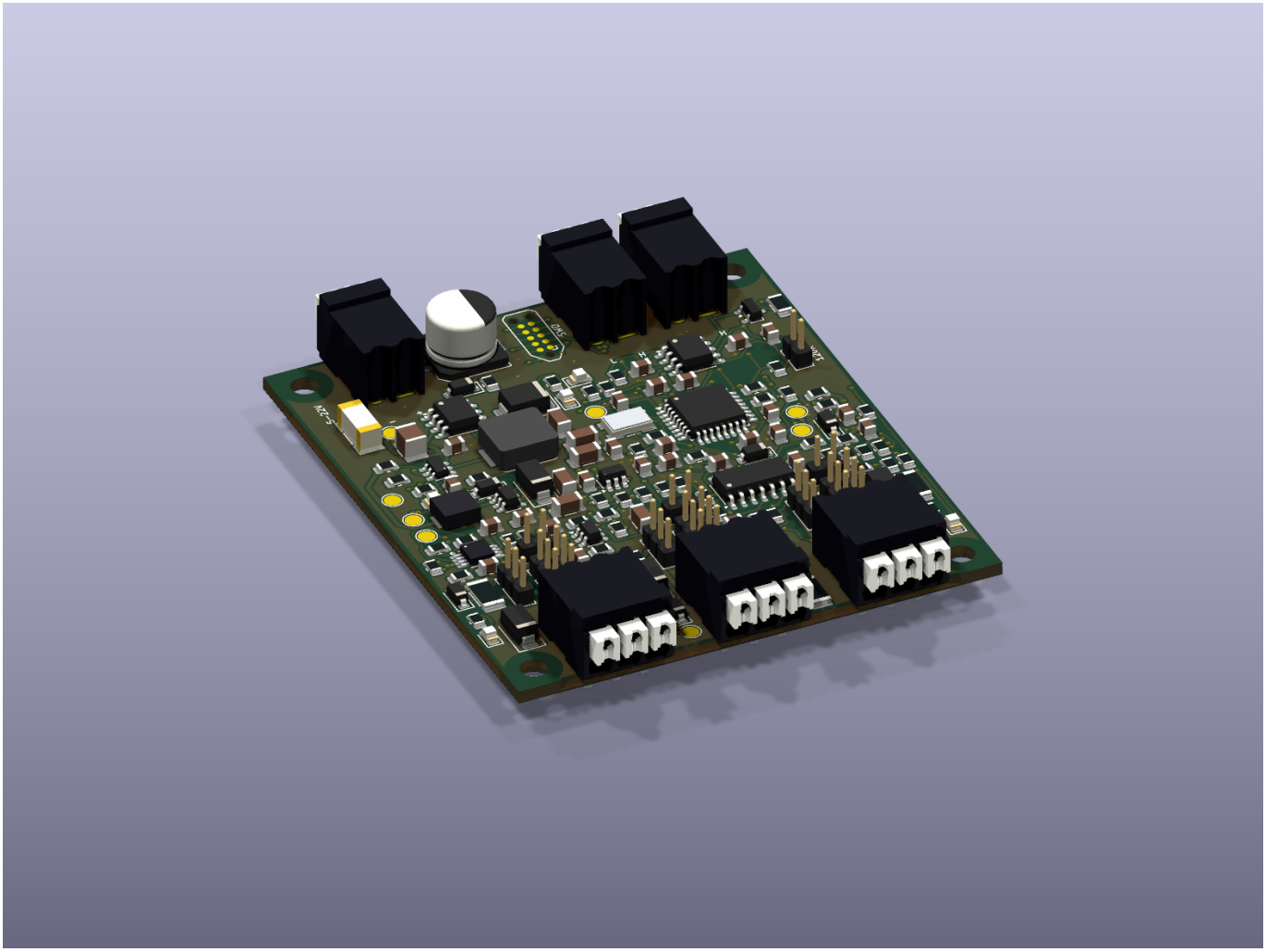
Source: docs/CONNECTORS.md

Scope: Rev B. Revision overview · All revisions.

Board references	Manufacturer order code	Full designation	DigiKey code
J1, J2, J3	1824420000	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4, J5, J6	1824430000	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Both are Weidmüller OMNIMATE LSF-SMT 3.50 push-in terminals with white push buttons and black housings. `/90` identifies horizontal wire entry parallel to the PCB. `/135` is angled and `/180` is vertical; those are not the current parts. The selected terminals have THT/THR solder legs, despite the series name containing “SMT”.

Manufacturer product pages and photographs: 2-pole 1824420000, 3-pole 1824430000. Local drawings: 2-pole datasheet, 3-pole datasheet.



Why the earlier render looked wrong

The earlier project models were simple dimensional boxes with guessed button shapes. The purchasing codes and horizontal selection were correct, but those envelopes did not depict the real housing, buttons or wire openings accurately.

The current board renders and mechanical STEP now use **official manufacturer STEP geometry** downloaded through Weidmüller's public Support Center. Original files are retained unchanged under `hardware/library/3d/vendor/`. A rigid translation places rear pin one at the footprint origin and the seating plane on the PCB. Display colors are assigned by the project to match the black housing, white actuators and metallic pins. The geometry has not been redrawn or rescaled.

The checked STEP pair provides 3.5 mm pole pitch, 7 mm pin-row spacing, two legs per pole, 3.5 mm pin length and 8.5 mm height above the PCB. All ten unique modeled pin centers align with their footprint pads; the pin cross-section fits within the nominal 1.1 mm drill. This is a CAD check, not a measured production-part fit test. The existing native PCB, connector footprints and BOM codes are unchanged.

The vendor CAD's body has a small nominal offset from the generic drawing; its transformed geometry fits inside the existing footprint courtyard and the 14.75 mm reserved depth. Continue reserving **60 × 76.1 mm**, plus wires and actuator access, for the complete board assembly. Supplier CAD is nominal and is not a substitute for checking actual samples and tolerances.

Reproduce and inspect

Run `python scripts/import_connector_models.py` in this revision directory with its Python environment activated. The script verifies the pin pattern, drill fit and vertical envelope before exporting the aligned models. Provenance, hashes and pin coordinates record the exact source files, public URLs and translation.

Regenerate the PCB renders and assembly STEP after importing the models. Manufacturer names and CAD remain third-party material; the project's original envelope models are no longer the geometry used for these horizontal terminals.